THE LOCUS WHERE THE RELATIVE DIMENSION IS BOUNDED BY N IS OPEN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a presentation reduction; the definition of relative dimension at a point of an algebraic space is what makes the identification U_n = a^{-1}(W_n) immediate, and no step is left out.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Lifting a morphism of algebraic spaces to a diagram of schemes. Spaces, Lemma spaces-lemma-lift-morphism-presentations, used at proof lines 7018-7027. Produces U, V and the morphism h.
- The scheme case: the bounded relative dimension locus is open. Morphisms, Lemma morphisms-lemma-openness-bounded-dimension-fibres, used at proof lines 7028-7031. Gives openness of U_n in U.
- Relative dimension of a morphism of algebraic spaces at a point, defined through a presentation. used at proof lines 7031-7033. Identifies the preimage of W_n with U_n.
- The topology on |X| defined through the topology of a presentation. used at proof lines 7034-7035. Concludes that W_n is open.

Source reference. Original lemma 02FZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3210–3210. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02X1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3824–3824. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $n \geq 0$. Assume f is locally of finite type. The set*

$$W_n = \{x \in |X| \text{ such that the relative dimension of } f \text{ at } x \leq n\}$$

is open in $|X|$.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Choose a diagram

$$\begin{array}{ccc} U & \xrightarrow{h} & V \\ {\scriptstyle a}\downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

where U and V are schemes and the vertical arrows are surjective and étale, see Spaces, Lemma 02X1. By Morphisms, Lemma 02FZ the set U_n of points where h has relative dimension $\leq n$ is open in U. By our definition of relative dimension for morphisms of algebraic spaces at points we see that $U_n = a^{-1}(W_n)$. The lemma follows by definition of the topology on $|X|$. $\square$